

UNCONDITIONAL LIKELIHOOD TESTS FOR THE LINEAR MODEL FOLLOWING BOX-COX TRANSFORMATION*

R.A.J. Wixley: National Research Institute for Mathematical Sciences
of the CSIR, P.O. Box 395, Pretoria, 0001, South Africa

Key words: Box-Cox transformation; conditional tests; F-tests;
likelihood ratio; unconditional tests.

Summary: The conventional linear model tests after power transformation to normality are conditional on the estimated transformation parameter. Unconditional likelihood ratio tests are shown to have a more correct level, both theoretically and by means of a simulation study.

1. INTRODUCTION

This paper examines the performance of unconditional likelihood ratio tests of hypotheses concerning linear model parameters after the standard Box-Cox transformation $z_{0i}^{(\lambda)}$, of observations $y_1, \dots, y_n$, where

$$(1) \quad z_{0i}^{(\lambda)} = \begin{cases} \frac{y_i^\lambda - 1}{\lambda} & , \quad \lambda \neq 0 \\ \ln y_i & , \quad \lambda = 0 \end{cases}$$

* This paper is based on part of an M.Sc. thesis presented to the University of South Africa, Pretoria, Republic of South Africa.

and $z_{0i}^{(\lambda)} = \underline{x}_i' \underline{\beta}_0 + \epsilon_i \sigma_0$ with $\epsilon_i \sim N(0,1)$,

where $\underline{\beta}_0$ and $\underline{x}_i$ are vectors length p and n respectively with $\beta_{01} = \mu_0$, a location parameter. Without loss of generality, we may take X to be of full column-rank where $X' = (\underline{x}_1, \dots, \underline{x}_n)$, and $\underline{x}_1 = 1$.

Box and Cox (1964, 1982) and Hinkley and Runger (1984) argued that inferences on the linear model should be "conditional on the scale" selected. A contrary view was held by Bickel and Doksum (1981), Carroll (1982) and Boksum and Wong (1983).

If the transformation is chosen as originally advocated by Box and Cox (1964), on the basis of an inspection of the profile likelihood λ and pragmatic consideration with a view to the sensible scientific interpretation of the data, then the conditional inference on the scale thus chosen presents no new problems. If the contrary view of the Box-Cox model is taken, one is led to consider the level and power of unconditional tests of hypotheses on the linear model. In this latter case, the "conventional" procedure is to estimate λ by maximum likelihood and apply the usual F-test in the transformed scale. This F-test may be regarded as a "conditional" likelihood ratio test.

Carroll (1982) showed in a two-group analysis of covariance model, that the conventional test of the group contrast had the correct level provided "the design is balanced over the two treatments in the first two moments of $\{x_i\}$ " (the covariate). Doksum and Wong (1983) proved that "two-sample and linear regression t-tests after transformation have asymptotically the correct level and asymptotically the same local power as if the statistician knew the transformation beforehand". They showed for the two-sample case that there is a substantial gain in power when the Box-Cox model is true, but that the transformed t-test may actually be worse than the untransformed test when the data is from exponential, or gamma distribution models.

In section 2 the unconditional test criteria are derived, and in section 3 the results of an extensive simulation study are presented. The results show that Box and Cox (op. cit.) and Hinkley and Ringer (op. cit.) are right.

The unconditional tests are more complicated than the conventional test, and not much better.

For convenience the discussion is conducted in terms of the normalized Box-Cox model

$$(2) \quad z_{1i}^{(\lambda)} = \begin{cases} \frac{y_i^\lambda - 1}{\lambda y_i^{\lambda-1}} & , \quad \lambda \neq 0 \\ y_i \ln y_i & , \quad \lambda = 0, \end{cases}$$

with $z_{1i}^{(\lambda)} = \underline{x}_{i1}' \underline{\beta}_1 + \epsilon_i \sigma_1$ with assumed $\epsilon_i \sim N(0,1)$ as before,

$$\dot{y} = \prod_{i=1}^n y_i^{1/n} \quad \text{and} \quad \underline{\beta}_1 = (\mu_1, \Gamma_{11}, \dots, \Gamma_{1,p-1}) .$$

2. DIRECT LIKELIHOOD INFERENCE AND THE BOX-COX MODEL

"Direct likelihood inference", a term coined by Aitkin (1982), refers to inference made directly from the relative likelihood function. If $\ell(\hat{\theta}_1, \hat{\theta}_2, \dots)$ is a multiparameter likelihood, with maximum likelihood estimators $(\hat{\theta}_1, \hat{\theta}_2, \dots)$, the relative profile likelihood of, say θ_1 is defined as

$$PR(\theta_1) = \ell(\theta_1; \tilde{\theta}_2, \tilde{\theta}_3, \dots) / \ell(\hat{\theta}_1, \hat{\theta}_2, \dots),$$

where $\tilde{\theta}_2, \tilde{\theta}_3, \dots$ satisfy $\frac{\partial \ell}{\partial \theta_2} = \frac{\partial \ell}{\partial \theta_3} = \dots = 0$.

For the linear model parameters $\underline{\beta}_1$

$$\ell(\tilde{x}_1, \underline{\beta}_1, \tilde{\sigma}_1) = \left[(z_1^{(\tilde{\lambda})} - \underline{x}_{\underline{\beta}_1})' z_1^{(\tilde{\lambda})} - \tilde{x}_{\underline{\beta}_1} \right]^{-\frac{n}{2}},$$

and the relative profile likelihood of $\underline{\beta}_1$ is

$$PR(\underline{\beta}_1) = \left[\frac{(z_1^{(\tilde{\lambda})} - \underline{x}_{\underline{\beta}_1})' (z_1^{(\tilde{\lambda})} - \underline{x}_{\underline{\beta}_1})}{z_1^{(\hat{\lambda})'} A_{z_1}^{-1} z_1^{(\hat{\lambda})}} \right]^{-\frac{n}{2}},$$

where $A = I - X(X'X)^{-1}X'$.

Suppose we wish to test the hypothesis

$$(3) \quad H_0 : \beta_1 = \beta_1^* = (\mu_1, \Gamma_{11}, \dots, \Gamma_{1,p-v_1-1}, 0, \dots, 0)' = (\beta_{11}^* : 0')$$

in which v_1 of the effects Γ_{1j} are set to zero, corresponding to the model

$$E(z_1^{(\lambda)}) = X\beta_1^* = X_1\beta_{11} + X_2 0.$$

The conventional F-test F^{CONV} , follows from the conditional likelihood ratio test Δ_c since

$$(4) \quad \Delta_c = \frac{\left(\frac{z_1^{(\hat{\lambda})'} A^* z_1^{(\hat{\lambda})}}{z_1^{(\hat{\lambda})'} A z_1^{(\hat{\lambda})}} \right)^{-\frac{n}{2}}}{\left(1 + \frac{v_1}{v_2} F^{\text{CONV}} \right)^{-\frac{n}{2}}},$$

where $A^* = I - X_1(X_1'X_1)^{-1}X_1'$.

The likelihood ratio criterion for the hypothesis (3) may be shown after some algebraic manipulation to be

$$(5) \quad \Delta = \frac{\left(\frac{z_1^{(\tilde{\lambda})'} A^* z_1^{(\tilde{\lambda})}}{z_1^{(\hat{\lambda})'} A z_1^{(\hat{\lambda})}} \right)^{-\frac{n}{2}}}{\left(1 + \frac{v_1}{v_2} F^{\text{CONV}} + \frac{\tilde{\epsilon}^{*'} \tilde{\epsilon}^* - (z_1^{(\tilde{\lambda})} - X\tilde{\beta}_1^*)' (z_1^{(\hat{\lambda})} - X\beta_1^*)}{z_1^{(\hat{\lambda})'} A z_1^{(\hat{\lambda})}} \right)^{-\frac{n}{2}}}$$

$$= \left(1 + \frac{v_1}{v_2} F^{\text{LR}} \right)^{-\frac{n}{2}},$$

where $\tilde{\epsilon}^* = z_1^{(\tilde{\lambda})} - X\tilde{\beta}_1^*$.

In (5) above F^{CONV} and F^{LR} are the "conventional" and "likelihood ratio" F statistics with (v_1, v_2) degrees of freedom, with v_1 defined above, and $v_2 = r(A)$. Clearly, $F^{\text{CONV}} > F^{\text{LR}}$ since $\|\epsilon^*\|^2 \leq \|\tilde{z}_1^{(\tilde{\lambda})} - \tilde{x}\tilde{\beta}^*\|^2$.

As a consequence the test based on F^{LR} may have a more correct level than F^{CONV} , the level of which may be expected to be slightly inflated, particularly when the original observations y are normally distributed.

Alternatively, we have the classical likelihood ratio test of H_0 :

$$(6) \quad -2 \ln \Delta = n \left(\ln \left(\tilde{z}_1^{(\tilde{\lambda})'} A^* \tilde{z}_1^{(\tilde{\lambda})} \right) - \ln \left(\tilde{z}_1^{(\hat{\lambda})'} A \tilde{z}_1^{(\hat{\lambda})} \right) \right).$$

This statistic is asymptotically distributed $\chi_{v_1}^2$ when H_0 is true. To improve the performance of this test statistic when v_1 is small, we consider two modifications. Firstly Kendall and Stuart (1967, vol. II, p. 242) remark that a good principle in constructing a likelihood ratio (LR) test is to adjust all maximum likelihood estimators used in the process so that they are unbiased. In the present case this yields

$$(7) \quad -2 \ln \Delta^*(\text{adj}) = n \left(\ln \left(\tilde{z}_1^{(\tilde{\lambda})'} A^* \tilde{z}_1^{(\tilde{\lambda})} / v_2^* \right) - \ln \left(\tilde{z}_1^{(\hat{\lambda})'} A \tilde{z}_1^{(\hat{\lambda})} / v_2 \right) \right) \\ = n \ln(s_1^2/s_0^2), \quad \text{where } v_2^* = v_2 + v_1.$$

For the second modification, it is easy in the present case to obtain an approximate Bartlett adjustment to the LR test above so that if

$$w = c(-2 \ln \Delta),$$

we may find c such that

$$E(w) = E(\chi_{v_1}^2) = v_1.$$

We know that if the null hypothesis H_0 is true, then

$$\tilde{z}_1^{(\tilde{\lambda})'} A^* \tilde{z}_1^{(\tilde{\lambda})} \frac{1}{2\sigma^2} \sim \text{gamma} \left(1, \frac{v_2^*}{2}, 0 \right) \text{ approximately,}$$

and that

$$z_1^{(\hat{\lambda})} \sim A z_1^{(\hat{\lambda})} \frac{1}{2\sigma^2} \sim \text{gamma} \left(1, \frac{v_2}{2}, 0 \right) \text{ approximately.}$$

In the case of 'b' randomized blocks of 't' treatments, $v_2^* = n-b$, and $v_2 = n-b-t+1$. If x has a standard gamma distribution with parameters $(1, p, 0)$, then, as given by Kendall and Stuart (1965, p. 243, Eqn. 24.42),

$$E\{\ln ax\} = \ln a + \ln p - \frac{1}{2p} - \frac{1}{12p^2} + O\left(\frac{1}{p^3}\right).$$

Hence, disregarding the approximations made in assuming the gamma distribution above, we obtain

$$\begin{aligned} E(-2 \ln \Delta) &= n \left(\ln \frac{2\sigma^2}{n} + \ln \left(\frac{n-b}{2} \right) - \frac{1}{n-b} - \frac{1}{3(n-b)^2} + O\left(\frac{1}{p^3}\right) \right) \\ &\quad - n \left(\ln \left(\frac{2\sigma^2}{n} \right) + \ln \left(\frac{n-b-t+1}{2} \right) - \frac{1}{n-b-t+1} - \frac{1}{3(n-b-t+1)^2} + O\left(\frac{1}{p^3}\right) \right) \\ &= (t-1) \frac{n}{n-b} \left(1 + \frac{1}{2} \left(\frac{t-1}{n-b} + \frac{2}{n-b-t+1} \right) + \frac{1}{3} \left(\left(\frac{t-1}{n-b} \right)^2 + \frac{2}{(n-b-t+1)^2} \right. \right. \\ &\quad \left. \left. + \frac{1-t}{(n-b-t+1)(n-b)} \right) \right) + O\left(\frac{1}{n^3}\right) \\ &= (t-1) \frac{n}{n-b} \left(1 + \frac{1}{2} \left(\frac{t-1}{n-b} \right) + \frac{1}{n-b-t+1} + O\left(\frac{1}{n^2}\right) \right) \end{aligned}$$

so that we may use

$$(8) \quad c = \frac{n-b}{n} \left(1 + \frac{1}{2} \left(\frac{t-1}{n-b} \right) + \frac{1}{n-b-t+1} \right)^{-1}$$

to adjust $(-2 \ln \Delta)$ so that its expectation coincides more closely with $(t-1) = v_1$.

The performance of these tests is evaluated in a simulation study in the next section.

3. SIMULATION STUDY

3.1 Experimental design

This study investigates the level and power of the conventional F-test after transformation (4), and its likelihood ratio alternatives (5), (6), (7) and (8), for the specific situation of a randomised block design with six treatments and varying numbers of blocks. The study is in two parts: in the first the power and level of the various F-statistics are investigated, while the second part concerns the level of all tests proposed.

In the first part four factors were varied in a $(2 \times 2 \times 3 \times 4)$ factorial simulation experiment, each trial being based on 1 000 simulations of its particular model. The factors varied were the general mean, number of blocks, size of treatment effect, and the transformation parameter in the model

$$\omega_{ij}^{(\lambda)} = \mu_i + B_j + T_j + \sigma e_{ij}.$$

Experimental factors: (Part 1)

(Standard error $\sigma = 2,1$)

- | | |
|---------------------------------|---|
| 1. General mean (M) | (a) 13,5
(b) 23,5 |
| 2. Block effects (N) | (a) 4 blocks (-2,2,-1,1)
(b) 6 blocks (-2,2,-1,1,0,0) |
| 3. Treatment effects (T) | (a) $(-7,-3,1,1,3,9) \times \frac{1}{4}$
(b) $(-7,-3,1,1,3,9) \times \frac{1}{8}$
(c) (0, 0,0,0,0,0) |
| 4. Transformation parameter (L) | (a) 1
(b) 0,5
(c) 0
(d) -1 |

The error in Part 1 ($\sigma = 2,1$) was chosen to give a wide range of powers in the resulting tests. In Part 2 we investigate the level of the tests and, in particular, the asymptotic behaviour of the likelihood ratio test. Here only the number of blocks (N) and the transformation parameter (L) have been varied in a (5×4) simulation experiment with 1 600 simulations per trial. This number of simulations gives approximately 95% confidence intervals of $0,05 \pm 0,01$ for the $\alpha = 0,95$ level. (The choice of $\sigma = 0,5$ is somewhat arbitrary, the reasons are historical since the results given here formed part of a larger simulation study.)

Experimental factors: (Part 2)

($\mu = 13,5$, $T = (c)$ zero effects); $\sigma = 0,5$)

1. Number of blocks (N) in the series $(-1,1,0,0,-1,1,0,0,\dots)$

- (a) 4
- (b) 8
- (c) 16
- (d) 32
- (e) 64

2. Transformation parameter (L)

- (a) 1
- (b) 0,5
- (c) 0
- (d) -1

3.2 Simulation method

The objective here is to have sets of generated data for a range of λ 's, which, when transformed, correctly possess the identical linear model.

We have used the normalized simple transformation to generate the data. The attractive feature of this transformation is that its inverse is itself

a normalized simple transformation and the range of generated data may be strictly controlled. Also, when now applying the Box-Cox model to data generated in this way, the only parameter estimate that differs in expectation to the parent values is the location parameter, which is modified by the correction for continuity.

- (1) Variates were generated according to the model

$$z_{ij} = \mu + \Gamma_i + \beta_j + \sigma e_{ij} ,$$

where e_{ij} is a pseudo-random normal deviate generated by the IMSL subroutine GGNML.

- (2) The geometric mean of the expected values

$$\dot{g} = \prod_{i,j} (\mu + \Gamma_i + \beta_j)^{1/n}$$

was obtained.

- (3) The inverse simple transformation, using the geometric mean of the expectations was applied, giving

$$y_{ij} = \begin{cases} \frac{\frac{x_{ij}^{\frac{1}{\lambda}}}{\dot{g}^{\frac{1}{\lambda}}} - 1}{\frac{1}{\lambda}} & , \quad \lambda \neq 0 \\ \exp\left(\frac{x_{ij}}{\dot{g}}\right) & , \quad \lambda = 0 \end{cases}$$

Because of the discontinuity at $\lambda = 0$ for the simple transformation, the simulation data for $\lambda = 0$ were generated by an exponential transformation, as shown above, and are thus in the Box-Cox family, not the simple family. They were thus not on the same scale as the data for $\lambda = 1, 0,5$ and -1 .

Throughout, we have used a nominal test level of $\alpha = 0,05$, but because the exact critical values of the various tests are, with one exception, unknown, it is not possible to make direct power comparisons.

The tests considered are:

(1) *Conventional F-tests*

- (a) Untransformed (i.e. $\lambda = 1$), with degrees of freedom $v_1 =$ no. treatments - 1 = 5, $v_2 = 5 \times (\text{no. blocks} - 1)$;
- (b) data transformed by maximum likelihood $\hat{\lambda}$, with degrees of freedom (v_1, v_2) (F^{CONV} , eqn (4));
- (c) as for (b) but with degrees of freedom $(v_1, v_2 - 1)$.

(2) *Likelihood ratio F-tests*

- (a) With (v_1, v_2) degrees of freedom (F^{LR} , eqn (5));
- (b) with $(v_1, v_2 - 1)$ degrees of freedom.

(3) *Likelihood ratio tests (LR)*

- (a) Wald's likelihood ratio criterion with v_1 degrees of freedom (eqn (6));
- (b) as for (a), but with quadratic forms or maximum likelihood estimates replaced by unbiased estimates of variance (eqn (7)), i.e. $n \ln(s_1^2/s_0^2)$.
- (c) Wald's LR criterion adjusted to have the correct expectation, i.e. $c(-2 \ln \Delta)$ where c is given in eqn (8).

3.3 Results

3.3.1 *Part 1: Power and Level of F-tests*

Table (1) gives the relative frequencies P of significant F-tests at the 5% nominal level in 1 000 simulations. To avoid needless repetition, only

those simulations with $N = 4$ blocks are given.

Approximate values for the power of the exact F-test (as interpolated from the Biometrika, vol. 38 (1951) tables of Pearson and Hartley) were obtained for two of the linear models considered and agreement with the observed powers was generally good. Because these relative frequencies "P" are binomial variates, they have been transformed to the $\sin^{-1}\sqrt{P}$ scale before being subjected to the analysis of variance, which showed significant L effects only for the conventional untransformed test 1(a). For all tests following transformation, the L effects were not significant. Table 2 gives the results for tests 1(a), 1(b) and 2(b).

When $\lambda = 1$, test 1(a) is exact. For this reason, in table 3 we show contrasts in the $\sin^{-1}\sqrt{P}$ scale between tests 1(a) and tests 1(b), 1(c), 2(a) and 2(c) for those simulations with $\lambda = 1$. Test 2(b) has the closest level to the exact test, with an overall mean contrast of -0,0051 radians. These contrasts were unaffected by factors M, T and N.

Pairwise contrasts between the four tests 1(b), 1(c), 2(a) and 2(b) were analysed using all simulations. In no case was the factor L significant, but the factors T and N were significant for most contrasts. The mean contrasts were in each case highly significantly different from zero, as shown in table 4. Note that the comparison 1(c)-2(a) which had the smallest contrast, showed no significant effects in the analysis of variance. The difference between these two tests is thus parsimoniously summarized by the overall mean contrast of $0,0015 \pm 0,0005$ radians, which corresponds to a difference in probability of 0,0007 at $P = 0,05$ and ,0015 at $P = 0,5$. The mean contrast of 0,0077 radians between 1(b) and 2(a) is almost identical to the mean contrast of 0,008 between 1(c) and 2(b). These contrasts correspond to comparisons between F^{CONV} and F^{LR} , respectively without and with a degree of freedom adjustment for estimation of λ . These results are thus in accord with the theory of section 2, which suggested that the unconditional test F^{LR} should have a more correct level than the conventional test F^{CONV} .

3.3.2 Part 2: Level of tests

Table (5) shows the results of the study of test level. The likelihood ratio test LR 3(c), which is adjusted to have the correct expectation, has an observed level that is intermediate between the conventional F-test 1(b) and the likelihood ratio F-test 2(a). The LR test 3(a) converges with increasing N to an inflated level around 0.1 that appears to coincide with that achieved by the modified LR test 3(b). These results are clearly illustrated in figure 1. The interest of these results lies in the demonstration of just how severe the bias of the unadjusted Wald criterion can be. When, in addition, the sample size is small, it is clear that in routine application, the unadjusted likelihood ratio test 2(a) can be a dangerously imprecise implement.

4. CONCLUSIONS

Conditional and unconditional tests of hypotheses concerning linear model after Box-Cox transformation to normality were compared in a simulation study and were found to have almost identical power characteristics.

The level of the conventional F-test following transformation, with no adjustment in degrees of freedom to allow for transformation, is slightly inflated relative to the level of the unconditional likelihood ratio F-test, with the same degrees of freedom. The latter has the more correct level. The difference is small and of no practical importance. Adjustment of the conventional F-test by subtracting a single degree of freedom from the denominator results a more correct level and can be recommended as a routine practice.

Of the unconditional likelihood ratio test the likelihood ratio F-test with a single degree of freedom adjustment to the numerator appears to have the most correct level. The "Bartlett" adjusted likelihood ratio criterion has a satisfactory performance, but the unadjusted likelihood ratio criterion is dangerously imprecise.

The investigation has been confined to randomized block arrangements where it is known that F-test procedures provide a practical approximation to randomization tests (Kempthorne 1952).

Extrapolation of the simulation results to a wider class of regression models is not possible without further investigation. In any event, this work provides unconditional tests that may be applied in any situation.

REFERENCES

- AITKEN, M. (1982). Direct likelihood inference. In GLIM 82: *Proceedings of the International Conference on Generalised Linear Models*. Ed. R. Gilchrist, New York, Springer Verlag, 76-86.
- BOX, G.E.P. & COX, D.R. (1964). An analysis of transformations. *J. Royal Statist. Soc. B*, **143**, 383-430.
- BOX, G.E.P. & COX, D.R. (1982). An analysis of transformations revisited, rebutted. *J. Amer. Statist. Ass.*, **7**, 209-210.
- BICKEL, P.J. & DOKSUM, K.A. (1981). An analysis of transformations revisited. *J. Amer. Statist. Ass.*, **76**, 296-311.
- CARROLL, R.J. (1982). Tests for regression parameters in power transformation models. *Scand. J. Statist.*, **9**, 217-222.
- DOKSUM, K.A. & CHI-WING WONG (1983). Statistical tests based on transformed data. *J. Amer. Statist. Ass.*, **78**, 411-417.
- KEMPTHORNE, O. (1952). *The design and analysis of experiments*. John Wiley & Sons, New York.
- KENDALL, M.G. & STUART, A. (1967). *The advanced theory of statistics*, vol. II. Charles Griffin, London.
- WIXLEY, R.A.J. (1984). *Transformations and inference*. M.Sc. thesis, University of South Africa, Pretoria.

TABLE 1

PERFORMANCE OF F-TESTS: RELATIVE FREQUENCY OF SIGNIFICANT
F-TESTS AT 5% NOMINAL LEVEL IN 1 000 SIMULATIONS
(RESULTS GIVEN FOR N = 4 BLOCKS ONLY)

μ	T	λ	*conventional			likelihood		Approximate power of exact F-test with 95% limits
			1(a)	1(b)	1(c)	2(a)	2(b)	
13,5	a	-1,0	0,317	0,451	0,435	0,430	0,422	
	b		0,084	0,128	0,121	0,120	0,115	
	c		0,046	0,064	0,063	0,062	0,061	
23,5	a		0,418	0,483	0,472	0,468	0,452	
	b		0,110	0,132	0,125	0,128	0,177	
	c		0,040	0,050	0,046	0,049	0,042	
13,5	a	0,0	0,434	0,455	0,439	0,440	0,431	
	b		0,107	0,119	0,116	0,114	0,112	
	c		0,047	0,052	0,047	0,048	0,044	
23,5	a		0,473	0,484	0,479	0,479	0,471	
	b		0,116	0,127	0,124	0,124	0,116	
	c		0,040	0,043	0,042	0,039	0,037	
13,5	b	0,5	0,432	0,455	0,442	0,442	0,434	
	c		0,140	0,147	0,139	0,137	0,133	
	d		0,057	0,064	0,063	0,063	0,060	
23,5	b		0,453	0,474	0,463	0,464	0,453	
	c		0,156	0,155	0,151	0,152	0,143	
	d		0,047	0,064	0,060	0,050	0,053	
13,5	b	1,0	0,454	0,483	0,473	0,472	0,460	0,42 $\pm$ 0,016
	c		0,132	0,155	0,147	0,145	0,139	-
	d		0,045	0,055	0,051	0,052	0,049	0,05 $\pm$ 0,013
23,5	b		0,441	0,465	0,457	0,457	0,449	0,42 $\pm$ 0,016
	c		0,127	0,138	0,130	0,133	0,127	-
	d		0,039	0,047	0,046	0,045	0,044	0,05 $\pm$ 0,013

NOTE: *the various tests are defined in paragraph 3.2.

LIKELIHOOD TESTS

TABLE 2

PERFORMANCE OF F-TESTS 1(a) AND 2(b) ASSESSED BY ANALYSIS OF VARIANCE IN $(\sin^{-1}\sqrt{p})$ SCALE, WHERE P IS THE PROPORTION OF SIGNIFICANT F-TESTS IN 1 000

Source	d.f.	Mean square and significance		
		conventional test		likelihood test
		1(a)	1(b)	2(b)
M	1	0,0027	0,0000	0,0001
T	2	1,608 ***	1,7113 ***	1,678 ***
L	3	0,0072 ***	0,0003	0,0003
N	1	0,1537 ***	0,1379 ***	0,1595 ***
M*T	2	0,0004	0,0001	0,0001
M*L	3	0,0020 **	0,0003	0,0003
M*N	1	0,0000	0,0001	0,0001
T*N	2	0,0578 ***	0,0575 ***	0,0593 ***
T*L	6	0,0016 **	0,0002	0,0002
N*L	3	0,0001	0,0003	0,0003
M*T*L	6	0,0007	0,0003	0,0003
error	17	0,00037	0,00035	0,00040

NOTE: *, **, *** denote significance at 5%, 1% and 0,1% levels respectively.

TABLE 3

PERFORMANCE OF F-TESTS WHEN $\lambda = 1$
 CONTRASTS IN $(\sin^{-1}\sqrt{p})$ SCALE BETWEEN THE EXACT TEST 1(a)
 AND TESTS 1(b), 1(c), 2(a), 2(b)

Contrast	overall mean	test of H_0 : zero mean
1(a)-1(b)	$-0,01925 \pm 0,0021$	***
1(a)-1(c)	$-0,01286 \pm 0,0016$	***
1(a)-2(a)	$-0,01175 \pm 0,0016$	***
1(a)-2(b)	$-0,0051 \pm 0,0015$	**

NOTE: *, **, *** denote significance at the 5%, 1% and 0,1% levels for a t-test with 11 degrees of freedom.

TABLE 4

MEAN PAIRWISE CONTRASTS BETWEEN F-TESTS 1(b), 1(c) and 2(b)
 ASSESSED BY ANALYSIS OF VARIANCE IN THE $(\sin^{-1}\sqrt{p})$ SCALE

	C o n t r a s t					
	1(b)-1(c)	1(b)-2(a)	1(b)-2(b)	1(c)-2(a)	1(c)-2(b)	2(a)-2(b)
mean	0,0062	0,0077	0,0142	0,0015	0,0080	0,0065
contrast	$\pm 0,0004$	$\pm 0,0006$	$\pm 0,0007$	$\pm 0,0005$	$\pm 0,0005$	$\pm 0,0004$
	***	***	***	**	***	***

NOTE: *, **, *** denotes significantly different from zero at 5%, 1%, and 0,1% levels respectively.

TABLE 5

MONTE CARLO STUDY OF THE LEVELS OF VARIOUS F AND LIKELIHOOD RATIO TESTS
 $M = 13,5$, $S = 0,5$. RELATIVE FREQUENCY OF SIGNIFICANCE OUT OF $n = 1\ 600$
 (NOMINAL LEVEL OF TESTS $\alpha = 0,05$ EXACT 95% LIMITS (0,0425, 0,06125))

no blocks	trans- formation par- ameter	F-tests				(3) likelihood ratio		
		(1) conventional (a)	(b)	(2) likelihood ratio (a)	(b)	(a)	(b)	(c)
N	λ	$\lambda=1(v_1, v_2) \cdot \hat{\lambda}:(v_1, v_2)$		$(v_1, v_2) \cdot (v_1, v_2-1)$		$n \ln \Delta$	$n \ln \left[\frac{s_1^2}{s_0^2} \right]$	corrected for bias
4	1,0	0,0475	0,0575	0,0538	0,0463	0,2056*	0,0894*	0,055
	0,5	0,0519	0,0631*	0,06	0,0575	0,1913*	0,0813*	0,0631*
	0,0	0,0513	0,0606	0,0594	0,0569	0,205 *	0,09 *	0,06
	-1,0	0,0388*	0,0531	0,0513	0,0506	0,1818*	0,0769*	0,0525
8	1,0	0,0469	0,05	0,0488	0,0475	0,14 *	0,0913*	0,0488
	0,5	0,0469	0,0525	0,0506	0,0488	0,1444*	0,0906*	0,056
	0,0	0,0438	0,0513	0,0494	0,0488	0,1431*	0,0913*	0,05
	-1,0	0,06	0,0688*	0,0656*	0,065 *	0,1344*	0,0869*	0,0663*
16	1,0	0,0519	0,0544	0,0525	0,0519	0,1225*	0,0969*	0,0525
	0,5	0,0425	0,0425	0,0419*	0,0419	0,1131*	0,0881*	0,0419*
	0,0	0,0544	0,0644*	0,0631*	0,0619*	0,14 *	0,1156*	0,0575
	-1,0	0,0481	0,0525	0,0519	0,0519	0,1094*	0,0881*	0,0519
32	1,0	0,0525	0,0544	0,0538	0,0538	0,1081*	0,1 *	0,0538
	0,5	0,0569	0,0581	0,0569	0,0569	0,0994*	0,085 *	0,0569
	0,0	0,0419*	0,0438	0,0425	0,0425	0,1113*	0,1006*	0,0425
	-1,0	0,0406*	0,0425	0,0413*	0,0413*	0,115 *	0,1044*	0,0413*
64	1,0	0,0419	0,0463	0,0463	0,0463	0,1038*	0,0975*	0,0463
	0,5	0,0513	0,0488	0,0488	0,0488	0,1094*	0,105 *	0,0488
	0,0	0,0594	0,0625	0,0625	0,0625	0,0994*	0,095 *	0,0625*
	-1,0	0,0481	0,0469	0,0456	0,0456	0,1056*	0,0963*	0,0456

NOTE: * denotes relative frequencies outside 95% nominal limits.

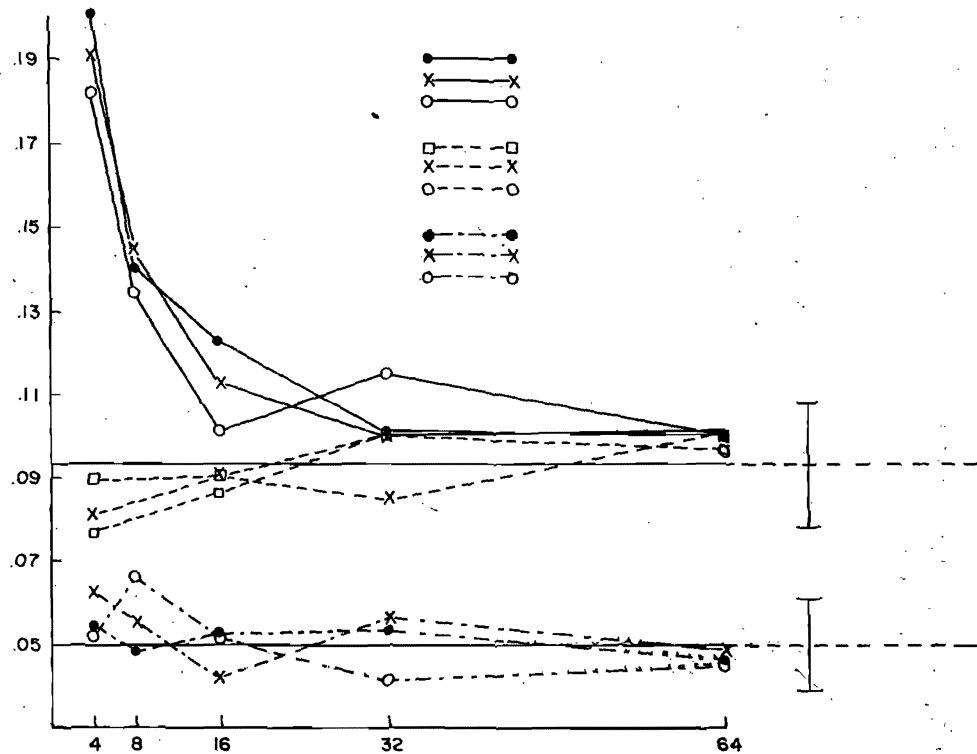


FIGURE 1. Simulation results part 2: level of likelihood ratio tests 3(a), (b) and (c) defined in section 3.2 (n = 1 600 simulations)